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United States Patent

12415299

Kind Code

B2

Date of Patent

September 16, 2025

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Mould for the production of closures in compression moulding machines

Abstract

A mould for the production of closures with tear-off membrane in a compression moulding machine, includes: a male die element including a first abutment surface and a female die element movable relative to each other along a moulding orientation. The mould is positionable between a closed configuration, in which the male die element and the female die element are in contact with each other, and an open configuration, in which the male die element and the female die element are spaced apart. The female die element includes: a first block, a second block and a third block, aligned with each other and movable along the moulding orientation; a first spring, interposed between the first block and the second block; a second spring, interposed between the second block and the third block and configured to apply an opening force along the moulding orientation. The first spring is configured to apply a closing force along the moulding orientation to keep the second block and the first block in contact with each other when no other forces are applied.

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Appl. No.: 18/405425

Filed: January 05, 2024

Prior Publication Data

Document Identifier

US 20240140005 A1

Publication Date

May. 02, 2024

Foreign Application Priority Data

IT 102019000017849 Oct. 03, 2019

Related U.S. Application Data

continuation parent-doc US 17754236 US 11904511 WO PCT/IB2020/059208 20201001 child-doc US 18405425

Publication Classification

Int. Cl.: **B29C43/36** (20060101); **B29C43/08** (20060101); **B29C43/50** (20060101); **B29C43/52** (20060101); **B29C43/58** (20060101); B29L31/56 (20060101)

U.S. Cl.:

CPC **B29C43/361** (20130101); **B29C43/08** (20130101); **B29C43/50** (20130101); **B29C43/52** (20130101); **B29C43/58** (20130101); B29C2043/3676 (20130101); B29C2043/5053 (20130101); B29L2031/565 (20130101)

Field of Classification Search

USPC: None

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. application Ser. No. 17/754,236, filed on 2022 Mar. 28, which was a national stage entry of PCT/IB2020/059208, filed on 2020 Oct. 1, that claims the benefit of Italian Application No. 102019000017849, filed 2019 Oct. 3, the entire disclosures of which are hereby incorporated herein by reference.

TECHNICAL FIELD

- (1) This invention relates to a mould, a machine and a method for the compression moulding of plastic closures.
- (2) Compression moulding machines comprise a rotary carousel which houses a plurality of compression stations, each disposed at a respective angular coordinate of the carousel and configured to perform a respective work cycle in one full turn of the carousel.
- (3) Each compression station comprises a mould, including a male die element (generally called “male”) and a female die element (generally called “female”). In a compression moulding machine, a charge of plastic material is first of all placed in the female die element. Next, the female die element slides relative to the male die element, coming into abutment against it in an abutment zone, to define a cavity in which the plastic charge is pressed so it spreads and conforms to the shape of the cavity to form the closure.
- (4) This said, it should be noted that this disclosure relates to a specific type of closure, used on bottles containing oil. These closures comprise a membrane to preserve the quality of the oil until the moment it is used and a tab connected to the membrane to allow the membrane to be removed. The oil container can be opened by pulling the tab to tear off the membrane so that the oil can be poured out of the container.

BACKGROUND ART

- (5) Prior art solutions for making closures of this kind are described, for example, in document CN106738559A, where the female die element (or “female”) is divided into blocks that are slidable relative to each other.
- (6) These solutions do not, however, solve some of the drawbacks due to the nature of the closure being made. One of the drawbacks that has not been completely overcome is that the plastic charge may not be spread uniformly.
- (7) In effect, the limited thickness of the membrane, corresponding to a very thin gap, makes it difficult for the plastic to flow through the passage afforded by the gap. In addition, the presence of the tab makes the common operations needed to extract the closure from the machine more

difficult, resulting, for example, in the membrane being detached in advance.

DISCLOSURE OF THE INVENTION

(8) This invention has for an aim to provide a mould, a machine and a method for the compression moulding of closures to overcome the above mentioned drawbacks of the prior art.

(9) This aim is fully achieved by the mould, the machine and the method of this disclosure as characterized in the appended claims.

(10) The mould according to this invention comprises a male die element and a female die element which move relative to each other to define an expansion chamber in which a charge of material is compressed into a predetermined shape. In this disclosure, the closure made by the mould is a closure with a tear-off membrane. Closures with tear-off membrane have two specific, additional features compared to standard closures: the presence of a very thin membrane and the presence of a tab used to tear the membrane off to allow the fluid to be poured out of the container on which the closure is applied.

(11) Since both the membrane and the tab have very narrow cross sections, the resistance to the migration of the charge of material during compression is high and may translate as non-uniform spreading of the charge in the expansion chamber.

(12) Moreover, the connecting point between the collar of the closure and the membrane must be very thin to allow the membrane to be torn off easily.

(13) Advantageously, this disclosure provides a mould in which the female die element comprises a first block, a second block and a third block, which are movable relative to each other along a moulding orientation. The mobility between the modules allows obtaining sequential closing and opening of the mould.

(14) The technical effect of this feature is that of defining the expansion chamber progressively, so that the compression process can start before the expansion chamber is completely defined. This step of compressing, which occurs before the expansion chamber is completely defined, enables the charge to reach even the outermost spaces (that is, the collar of the closure). To allow sequentially closing the first, second and third blocks, the mould of this disclosure comprises a first spring and/or a second spring.

(15) The first spring is configured to keep the first block and the second block spaced apart during opening and during closing. The second spring is configured to keep the second block and the third block spaced apart during opening and during closing.

(16) A machine for making closures is characterized by the possibility of changing the moulds as a function of the type of closure to be made. A machine can therefore be fitted with different moulds.

(17) The compression moulding machine may, for example, be a rotary machine. In such a case, the machine has a first platen, which holds the male part of the mould, and a second platen, which holds the female part of the mould.

(18) The distance between the first platen and the second platen is called “gape” and, in many cases, is not variable.

(19) Conventional moulds have a maximum extension along the moulding orientation that is smaller than that of the mould of this disclosure because they comprise single blocks that are more compact.

(20) It is therefore necessary to reduce the extension of the mould to enable it to be fitted also on a machine originally designed for conventional moulds.

(21) In this regard, this disclosure provides a variant of the first spring which is configured to keep the first and the second block in contact with each other during closure but allows the first and the second block to be spaced apart during opening so as to facilitate releasing of the tab.

(22) This disclosure also affords an advantageous solution for cooling the male and female mould parts. More specifically, each block comprises a respective internal circuit with a refrigerant fluid flowing inside it. In an embodiment, these internal circuits may be mounted in series or in parallel. The operating conditions of the mould may change, however, and in a given configuration in series

or in parallel, the cooling efficiency may not be adequate for one block or another. In this regard, this disclosure provides a reconfiguring device configured to vary a connecting configuration of the internal circuits of the first, second and third blocks between a configuration in series, in which the internal circuits of the first, second and third blocks are connected in series, and a configuration in parallel, in which the internal circuits of the first, second and third blocks are connected in parallel. (23) In an embodiment, this disclosure also provides a system for aiding closure release. In effect, with closures of this kind, the step of releasing is quite critical with regard to both the female and the male, which has to release more closure parts whose detachment is critical. To facilitate detachment, this disclosure provides a pressurizing duct which blows a jet of air through the tip of the male facing the female. This air jet produces a force that pushes the surface of the closure away from the male so that the closure can be withdrawn by an auxiliary unit.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) These and other features will become more apparent from the following detailed description of a preferred embodiment, illustrated by way of non-limiting example in the accompanying drawings, in which:
- (2) FIG. 1 schematically illustrates a compression moulding machine for making closures with tear-off membrane;
- (3) FIG. 1A schematically illustrates an embodiment of the machine of FIG. 1;
- (4) FIG. 2 is a schematic top view of the machine of FIG. 1;
- (5) FIG. 3 illustrates an embodiment of a closure with a tear-off membrane;
- (6) FIG. 4 shows a cross section of a mould for making a closure with a tear-off membrane;
- (7) FIG. 5 shows a detail of the cross section of FIG. 4 with the mould in a different operating configuration;
- (8) FIGS. 6A, 6B, 6C, 6D, 6E, 6F, 6G illustrate the detail of FIG. 4 with the mould in respective operating configurations;
- (9) FIGS. 7A and 7B illustrate, respectively, a first and a second configuration of a circuit for cooling a female die element of the mould of FIG. 4.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS OF THE INVENTION

- (10) With reference to the accompanying drawings, the numeral **1** denotes a compression moulding machine for making closures (that is to say, a machine for making closures by a compression moulding process).
- (11) According to an aspect of this disclosure, the machine **1** is configured to make a closure **100** used for closing oil containers.
- (12) The closure **100** comprises a body **101**. The closure **100** is axisymmetrically shaped about one of its axes. The body **101** comprises a first part **101A**, configured to be screwed to the cap of the oil container, and a second part **101B**, configured to be connected to the body of the oil container.
- (13) The first part **101A** of the closure body **101** comprises a first collar **101A'**, which extends from the second part **101B** along the axis of symmetry in an out direction U. The first part **101A** of the closure body **101** comprises a second collar **101A''**, which extends from the second part **101B** along the axis of symmetry in the out direction U. The first collar **101A'** and the second collar **101A''** are concentric. The first collar **101A'** surrounds the second collar **101A''**. The first collar **101A'** comprises an upper thread FS on an outside surface of it (relative to the axis of symmetry). The upper thread FS is configured to be screwed to a thread of a cap of the oil container.
- (14) The second part **101B** comprises a connecting collar **101B'** and a transverse wall **101B''**, perpendicular to the axis of symmetry.
- (15) The connecting collar **101B'** comprises a plurality of teeth **102** on an inside surface of it,

facing the axis of symmetry. The plurality of teeth **102** protrude from the inside surface of the connecting collar **101B'** to grip an edge of the oil container.

(16) The closure **100** comprises a circumferential sealing wall **103**, configured to create a fluid-tight seal between the closure **100** and the neck of the oil bottle the closure is mounted on. It should be noted that in this application, the circumferential sealing wall **103** is denoted more simply by the term “circumferential wall” **103**. The circumferential wall **103** extends from the transverse wall **101B''** along the axis of symmetry in the in direction E, opposite to the out direction U. The circumferential wall **103** is inclined to the axis of symmetry and converges towards the axis of symmetry in the out direction U.

(17) The transverse wall **101B''** comprises an opening A, through which the oil can flow out of the container.

(18) The closure **100** comprises a plurality of fins **104**. The plurality of fins are connected to the transverse wall **101B''**. The plurality of fins **104** extend from the transverse wall **101B''** along the axis of symmetry in the in direction E. The plurality of fins **104** is inclined to the axis of symmetry and converges towards the axis of symmetry in the in direction E, in such a way as to face the opening A (that is, to be at least partly aligned with the opening A along an orientation parallel to the axis of symmetry). The plurality of fins **104** allows conveying the oil and acts as an anti-bubbling element.

(19) The closure **100** comprises a membrane **105**. The membrane **105** is connected to the transverse wall **101B''** to close the opening A. The membrane **105** has a concave profile in the out direction to facilitate its removal.

(20) The closure **100** comprises a tab **106** (or removal element **106**). The tab **106** is connected to the membrane **105** to allow it to be removed when the oil container is opened.

(21) The machine **1** is therefore configured to make closures of this kind, which in the jargon of the trade, may be defined as “tear-off closures”.

(22) The machine **1** comprises a frame **1'**.

(23) The machine **1** comprises a rotary carousel **10**. The rotary carousel **10** is configured to rotate about a first axis of rotation R1. The rotary carousel **10** comprises an upper platen (disc, cylinder) **10A** and a lower platen (disc, cylinder) **10B**. The upper platen **10A** comprises an underside surface facing the second platen **10B**. The second platen **10B** comprises a top surface facing the first platen **10A**.

(24) The distance between the underside surface of the first platen **10A** and the top surface of the second platen **10B** is known as “gape” in the jargon of the trade and is labelled SB.

(25) The first platen **10A** comprises a plurality of upper housings **10A'**. In an embodiment, each upper housing **10A'** comprises a respective concavity, which opens onto the space between the first platen **10A** and the second platen **10B**. The housings **10A'** of the plurality are disposed on a circumference of the first platen **10A** and are spaced, preferably uniformly, around that outer circumference (that is to say, they are equispaced from each other).

(26) The second platen **10B** comprises a plurality of lower housings **10B'**. In an embodiment, each lower housing **10B'** comprises a respective concavity, which opens onto the space between the first platen **10A** and the second platen **10B**. The housings **10B'** of the plurality are disposed on a circumference of the second platen **10B** and are spaced, preferably uniformly, around that outer circumference (that is to say, they are equispaced from each other).

(27) Each housing **10B'** of the plurality of lower housings is aligned (at least partly) with a respective housing **10A'** of the plurality of upper housings, along an orientation parallel to the first axis of rotation R1.

(28) The machine **1** comprises a plurality of moulds **11**. Each mould comprises a male die element **2** (hereinafter denoted simply by the term “male”) and a female die element **3** (hereinafter denoted simply by the term “female”).

(29) Each mould **11** is movable between a closed configuration CC, in which the respective male **2**

and the respective female **3** are in contact to define an expansion chamber CE for expanding a (plastic) material and having a shape corresponding to the closure to be made, and an open configuration CA, in which the male **10A** and the female **10B** are spaced apart.

(30) Each male **2** of the plurality of moulds **11** is housed in a respective housing **10A'** of the plurality of upper housings. In an embodiment, each male **2** is fixed relative to the first platen **10A**. Each male **2** protrudes from the underside surface of the first platen **10A**.

(31) Each female **3** of the plurality of moulds **11** is housed in a respective housing **10B'** of the plurality of lower housings. Each female **3** protrudes from the top surface of the second platen **10B**. In an embodiment, each female **3** is movable relative to the second platen **10B** in such a way as to come into contact with the respective male **2** to define the expansion chamber CE.

(32) It should be noted that the movement of the female **3** relative to the male **2** is only one possible embodiment of this machine and in other embodiments, the male **2** might move relative to the female **3** or both the parts **2** and **3** might move towards and away from each other.

(33) The machine **1** comprises an actuating unit **12**. The actuating unit **12** is configured to move the parts of the machine **1**.

(34) The actuating unit **12** comprises a rotary actuator **121** configured to rotate the rotary carousel **10** about the first axis of rotation R1.

(35) For each full turn of the rotary carousel **10**, the machine **1** performs one working cycle in which it is configured to make a number of closures equal to the number of moulds **11** mounted on it. In other words, for each full turn of the rotary carousel **10**, each mould **11** makes one closure.

(36) As it rotates, the rotary carousel **10** defines one or more of the following working stations. a first working station SL1, in which each female **3** receives a charge of material and/or in which each male **2** releases a closure made in the preceding cycle (one complete turn of the rotary carousel **10**); a second working station SL2, in which the female **3** is moved towards and into contact with the male **2** to define the expansion chamber CE and to form the closure; the second working station SL2 being angularly spaced from the first working station SL1 by a closing angle AC; a third working station SL3, in which the female **3** is detached from the male **2** to allow the moulded closure to be released (or withdrawn); the third working station SL3 being angularly spaced from the first working station SL1 by a reopening angle AA.

(37) In an embodiment, the actuating unit **12** comprises a plurality of moulding actuators **122**. Each moulding actuator **122** is configured to move the corresponding female **3** (and/or the corresponding male **2**) along a moulding orientation S, parallel to the axis of rotation R1. In an embodiment, each moulding actuator **122** is configured to move the corresponding female **3** (and/or the corresponding male **2**) along the moulding orientation S in a closing direction VC, oriented from the female **3** to the male **2**, at the second working station SL2, to move the mould into the closed configuration CC. Each mould **11** remains in the closed configuration CC for angles of rotation of the rotary carousel included between the closing angle AC and the reopening angle AA.

(38) In an embodiment, each moulding actuator **122** is configured to move the corresponding female **3** (and/or the corresponding male **2**) along the moulding orientation S in an opening direction VA, oriented from the male **2** to the female **3**, at the third working station SL3, to move the mould into the open configuration CA. Each mould **11** remains in the open configuration CA for angles of rotation of the rotary carousel included between the reopening angle AA and the closing angle AC of the next cycle (the next turn).

(39) According to an aspect of this disclosure, embodiments are also imaginable which allow closing and reopening the mould: that is to say, to allow moving the mould between the closed configuration CC and the open configuration CA.

(40) In one of these embodiments, the movement of the mould between the closed configuration CC and the open configuration CA is performed by a transmission unit **123**, configured to transmit and/or convert the rotary motion of the rotary carousel **10** into a translational motion of a part of the mould **11**, preferably the female **3**.

(41) For example, for each female **3**, the machine **1** comprises a respective rod **123A** and a respective slide element **123B**. The rod **123A** is connected to the corresponding female at one end of it and to the slide element **123B** at the opposite end of it.

(42) The machine **1** comprises a guide cam **124** to which the frame **1'** is fixed. The guide cam **124** extends along a circumference which is aligned, along the moulding orientation **S**, with the circumference of the second platen **10B** which houses the females **3** of the moulds **11**.

(43) The slide element **123B** is in contact with the guide cam **124**. That way, the profile of the guide cam **124** defines, for each angular position of the rotary carousel **10**, a corresponding position of the female **3** along the moulding orientation **S**.

(44) The machine **1** comprises a control unit **13**, configured to send drive signals **131'** to the actuating unit **12** (or to the plurality of moulding actuators **122**, or to the rotary actuator **121**) in order to drive it. The machine **1** comprises a user interface **131**, connected to the control unit **13** to set working parameter values.

(45) The control unit **13** is configured to determine the drive signals **131'** as a function of the working parameter values entered by a user through the user interface **131**.

(46) Below is a detailed description of the features of each mould **11** of the plurality of moulds. For brevity, we will refer to a single mould, it being understood that one or more of the features described also apply to all the moulds **11** of the plurality.

(47) In an embodiment, the male **2** of the mould **11** comprises an outer bush. **21** (or extractor **21**). The outer bush **21** is configured to be coupled to the corresponding upper housing **10A'** of the first platen **10A**. The outer bush **21** comprises an abutment surface **21A**, which is configured to come into abutment against the female **3** in the closed configuration **CC** of the mould **11**. The abutment surface **21A** contributes to delimiting the expansion chamber **E**.

(48) In an embodiment, the male **2** comprises a toothed bush **22**. The toothed bush **22** is externally connected to the outer bush **21**. On its cylindrical outer surface, the toothed bush **22** comprises a plurality of recesses **22A**, which define the shape of the plurality of teeth **102** of the closure.

(49) In an embodiment, the male **2** comprises a receiving bush **23**. The receiving bush **23** is externally connected to the toothed bush **22**. The receiving bush **23** is externally connected to the toothed bush **22** in such a way as to leave, along a radial orientation perpendicular to the moulding orientation **S**, a space that defines a flow gap extending along the moulding orientation **S**.

(50) At the end of it facing towards the female **3**, the receiving bush **23** comprises a bevel **23A**. The bevel **23A** defines the inclination of the circumferential wall **103** of the closure.

(51) In an embodiment, the male **2** comprises a central block **24**. The central block **24** is externally connected to the receiving bush **23**. More specifically, the coupling between the receiving bush **23** and the central block **24** defines a circumferential gap **234** that is inclined to the moulding orientation **S**. This gap defines the shape and inclination of the plurality of fins **104**.

(52) The central block **24** comprises a membrane surface **241**, facing towards the female **3** and concave relative to the female **3** to define the concavity of the membrane **105** of the closure **100**.

(53) In an embodiment, the receiving bush **23** and/or the toothed bush **22** and/or the central block **24** are movable relative to the outer bush **21**. More specifically, in an embodiment, the unit including the receiving bush **23** and/or the toothed bush **22** and/or the central block **24** is movable relative to the outer bush **21** between a moulding position, in which the receiving bush is closer to the female **3** than the outer bush **21**, and a release position, in which the receiving bush is equidistant or further from the female **3** than the outer bush **21**, to allow the moulded closure **100** to be released.

(54) It should be noted that in a preferred embodiment, the unit including the receiving bush **23** and/or the toothed bush **22** and/or the central block **24** remains fixed relative to the frame while the outer bush **21** slides along the moulding orientation **S**. That way, the outer bush is configured to press down on the connecting collar **101B'** to detach the closure **100** from the male **2** so that the closure **100** can be withdrawn.

(55) In an embodiment of this disclosure, the central block **24** is movable relative to the receiving bush **23** along the moulding orientation S between a working position, in which the central block **24** is proximal to the receiving bush **23** to define the gap **234**, and a detached position, in which the central block **24** is distal from the receiving bush **23** to release the plurality of fins **104**.

(56) More specifically, the receiving bush **23** comprises a slide cavity **244** in which a piston of the central block **24** (that is, a piston attached to the central block) is received to slide therein.

(57) The slide cavity **244** defines an actuating chamber, located upstream of the piston of the central block **24** along the sliding orientation S in the closing direction VC. The male **2** comprises a pressurizing duct **243** (which, in one embodiment, is also the injection duct **251** of the first cooling circuit **25** of the male **2**). The pressurizing duct **243** is open onto the actuating chamber. Pressurizing the actuating chamber facilitates displacing the central block, thereby detaching the moulded closure. Displacement of the central block **24** is caused mainly by the displacement of the outer bush **21** (extractor), which is configured to push the closure **100** in the extraction direction VE. The displacement of the closure **100** results in a corresponding displacement of the central body **24**, which is entrained by the closure through the plurality of fins **104**.

(58) In an embodiment, the central block **24** is configured to slide in the slide cavity **244** without a fluid seal. In this embodiment, therefore, although it pushes the central block **24**, the air flows past the central block **24**.

(59) In an embodiment, the male **2** comprises at least one communication passage which connects the slide cavity **244** to the flow gap defined between the toothed bush **22** and the receiving bush **23**. In an embodiment, the communication passage is open onto the slide cavity **244** at a position opposite to the pressurizing duct **243** with respect to the central block **24**.

(60) In this embodiment, the air enters the slide cavity **244** through the pressurizing duct **243**, flows past the central block **24**, which is smaller in size than the slide cavity **244**, and reaches the communication passage and the flow gap to apply an ejecting force on the closure **100**.

(61) In an example embodiment, the central block **24** comprises a return spring **242**. The return spring **242** is configured to apply an elastic force on the central block **24** (that is, on the piston of the central block **24**) along the moulding orientation S in the opening direction VA. More specifically, the return spring **242** is disposed inside the slide cavity **244** at a position downstream of the piston of the central block **24** along the sliding orientation S in the closing direction VC. Thus, once the pressurizing duct **243** has pressurized the actuating chamber and the central block **24** has been taken to the detached position, depressurizing the actuating chamber combined with the action of the return spring **242** allows bringing the central block back to the working position.

(62) In an embodiment, the male **2** comprises a first cooling circuit **25**. The first cooling circuit **25** traverses the male **2** to remove heat therefrom. More specifically, in an embodiment, the cooling circuit **25** comprises an injection duct **251** configured to convey a cooling fluid to a zone proximate to the female **3**, —for example in an annular duct defined between the central block **24** and the receiving bush **23**. In an embodiment, the cooling circuit **25** comprises a recycling duct **252** configured to recycle the cooling fluid injected into the zone proximate to the female **3**, —for example by the annular duct.

(63) The female **3** comprises a first block **31**. In an embodiment, the female **3** comprises a second block **32**. In an embodiment, the female **3** comprises a third block **33**.

(64) The first block **31**, the second block **32** and the third block **33** are disposed one inside the other in a telescopic structure.

(65) In an embodiment, the first block **31** is movably connected to the second block **32**. In an embodiment, the first block **31** is connected to the second block **32** by a first guide element **311**. The first guide element **311** has a first end that is connected to the first block **31** and a second end that is connected to the second block **32**. The first guide element **311** is configured to slide in a respective seat **312** in the first block **31**. The seat **312** houses a first spring **313**, inside which the first guide element **311** passes.

(66) It should be noted that, in an embodiment, the spring is configured to apply, between the first block **31** and the second block **32**, a contact force having an orientation parallel to the moulding orientation S and a direction such as to move the two blocks closer together to keep them in contact with each other. In this embodiment, the first block **31** and the second block **32** are in contact with the mould in the open configuration CA. Further, when the first block **31** and the second block **32** are in contact, the first guide element **311** comes into abutment with the seat **312** in the closing direction VC. That means that in an embodiment, during the movement of the mould **11** from the open configuration CA to the closed configuration CC of the mould, the first block **31** and the second block **32** move as one. Otherwise, during the movement of the mould **11** from the closed configuration CC to the open configuration CA of the mould, the second block **32** is configured to slide relative to the first block **31** along the moulding orientation S in the opening direction until the first guide element **311** comes into abutment against the seat **312** in the opening direction.

(67) This therefore allows modular opening of the female **3**, where opening of the first block **31** is delayed with respect to the second block **32**.

(68) In other embodiments, the contact force is directed in such a way as to move the first block **31** away from the second block **32**. In such an embodiment, during the movement of the mould **11** from the open configuration CA to the closed configuration CC of the mould, the first block **31** is spaced from the second block **32**. The first block **31** comes into contact with the second block **32** after the first block **31** comes into abutment against the male **2**. In effect, the movement of the female **3** along the moulding orientation S, inhibited by the abutment between the first block **31** against the male **2**, causes the first spring **313** to be compressed, thereby displacing the second block **32** until it comes into contact with the first block **31**.

(69) In such an embodiment, therefore, also the closing of the second block **32** is delayed with respect to the first block **31**.

(70) In an embodiment, the second block **32** is movably connected to the third block **33**.

(71) The female **3** comprises a second spring **323**, interposed between the second block **32** and the third block **33** along the moulding orientation S.

(72) The female **3** comprises a second guide element **324**, which is elongate along the moulding orientation S and which is disposed inside the second spring **323** to guide the compression thereof.

(73) The guide element **324** is configured to stop against the second block **32** so as to limit the distance between the third block **33** and the second block **32** (that is, to limit the stretching of the second spring **323**). When the female **3** is opened, this feature allows the third block **33** to engage the second block **32** after moving (that is, sliding) for a certain length along the moulding orientation S until the guide element **324** stops against the second block **32** in the opening direction VA.

(74) The second spring **323** is configured to generate a respective contact force directed in such a way as to move the second block **32** away from the third block **33**. In such an embodiment, during the movement of the mould **11** from the open configuration CA to the closed configuration CC of the mould, the second block **32** is spaced from the third block **33**.

(75) The third block **33** comes into contact with the second block **32** after the second block **32** comes into abutment against the first block **31**. In effect, following contact between the second block **32** against the first block **31**, the movement of the female **3** along the moulding orientation S causes the second spring **323** to be compressed, thereby displacing the third block **33** until it comes into contact with the second block **32**.

(76) When the third block **33** has also stopped against the second block **32**, the expansion chamber is fully defined and the charge of plastic material is spread therein.

(77) This configuration of the female, with the first spring **313** and the second spring **323**, allows closing and opening the mould (moving between the open configuration CA and the closed configuration CC) in a modular fashion. These features offer several advantages. In effect, in these solutions, the material is not spread over the zone on the expansion chamber corresponding to the

membrane (whose thickness is very limited) until the other zones have been filled. The progressive closing therefore allows minimizing the time needed to spread the plastic material through the very narrow gaps, which might otherwise have a negative effect on the way the material is spread.

(78) In an embodiment, the first block **31** comprises an inner bush **314**. The inner bush **314** comprises an abutment surface, configured to come into abutment against the male **2**, preferably against the abutment surface **21A** of the outer bush **21** of the male **2**.

(79) In an embodiment, the inner bush **314** comprises a first cylindrical surface and a second cylindrical surface whose radius is smaller than that of the first cylindrical surface. In an embodiment, the first cylindrical surface is smooth. The first cylindrical surface delimits the expansion chamber CE and defines an outside surface of the connecting collar **101B'** of the closure **100**. The second cylindrical surface comprises a “female” thread which receives the charge of material and defines the upper thread FS of the closure **100**.

(80) In an embodiment, the second block **32** comprises a first bush **321** and a second bush **322**.

(81) The second bush **322** is disposed inside the first bush **321**. The first bush **321** and the second bush **322** are concentric. At the ends of them proximate to the male **2**, the first bush **321** and the second bush **322** are spaced apart to define a thickness of the second collar **101A''** of the closure **100**.

(82) The first bush **321** is concentric with the inner bush **314** of the first block **31**. In the closed configuration CC of the mould **11**, the first bush **321** and the second cylindrical surface of the inner bush **314** of the first block **31** are spaced apart to define a thickness of the first collar **101A'** of the closure **100**. The second bush **322** comprises a hollow in which the plastic material is configured to flow in order to form the tab **106** of the closure **100**.

(83) In an embodiment, the third block **33** comprises a plunger **331**, disposed inside the second bush **322** of the second block **32**. The plunger **331** is configured to slide inside the second bush **322** of the second block **32**.

(84) The plunger **331** comprises a convex surface which is configured to be coupled to the corresponding concave surface of the central block **24** of the male **2**. The convex surface of the plunger **331** and the concave surface of the central block **24** of the male **2** are, in the closed configuration CC of the mould **11**, spaced apart by a value that defines the thickness of the membrane **105** of the closure **100**.

(85) In an embodiment, the female **3** comprises a second cooling circuit **34**. The second cooling circuit **34** is configured to cool the female **3**, preferably the first block **31** and/or the second block **32** and/or the third block **33**. The second cooling circuit **34** comprises a recirculation duct **34'** configured to circulate a cooling fluid in a cooling direction. The recirculation duct traverses the first block **31** and/or the second block **32** and/or the third block **33**. More specifically, in an embodiment, the recirculation duct **34'** extends in the cooling direction, traversing first the third block **33**, then the second block **32** and, lastly, the first block **31**. This embodiment is purely exemplary since the order in which the modules are traversed may be varied, as anyone skilled in the art will readily understand.

(86) In an embodiment, the second cooling circuit comprises, for each first block **31**, second block **32** and third block **33**, one or more of the following features: a respective cooling inlet **341A**, **342A**, **343A**, through which the cooling fluid enters; a respective cooling outlet **341B**, **342B**, **343B**, through which the cooling fluid is conveyed out of the respective block; a respective internal duct **341**, **342**, **343**, each connected to a respective cooling inlet **341A**, **342A**, **343A** and to the respective cooling outlet **341B**, **342B**, **343B** to define a cooling path within the respective block.

(87) In an embodiment, the internal duct **343** of the third block comprises a first stretch, passing through the plunger **331** along the moulding orientation S in the closing direction VC, and/or a second stretch, passing through the plunger **331** along the moulding orientation S in the opening direction VA, and/or a third stretch which connects one end of the second stretch to the cooling outlet **343B**.

(88) In an embodiment, the internal duct **342** of the second block **32** comprises a cooling ring AR, formed on the outside of the first bush **321** of the second block **32** and connected to the cooling inlet **342A** and to the cooling outlet **342B** of the second block **32**.

(89) In an embodiment, the internal duct **341** of the first block **31** comprises a cooling chamber CR, formed on the outside of the inner bush **314** of the first block **31** and connected to the cooling inlet **341A** and to the cooling outlet **341B** of the first block **31**.

(90) In an embodiment, the internal ducts **341**, **342**, **343** of the first block **31**, second block **32** and third block **33** are connected to each other in series. In other embodiments, the internal ducts **341**, **342**, **343** of the first block **31**, second block **32** and third block **33** are connected to each other in parallel.

(91) According to an aspect of this disclosure, the actuating unit **12** is configured to move the female **3** (or the male **2**) along the moulding orientation at a variable speed. More specifically, the actuating unit **12** (that is, the moulding actuators **122**) is configured to move the female **3** (or the male **2**) along the moulding orientation at a speed that decreases as the female **3** approaches the male **2**. In addition, the actuating unit **12** (that is, the moulding actuators **122**) is configured to provide the female **3** (or the male **2**) with a force (torque) along the moulding orientation that increases as the female **3** approaches the male **2**.

(92) This feature allows providing more force when it is necessary to expand the material through the narrower gaps and, instead, a higher speed when the material flows more easily.

(93) In the same way, in the embodiment comprising the guide cam **124**, the profile of the cam is designed in such a way as to raise the female **3** rapidly during an initial step of moulding and to progressively reduce the slope of the profile as the female **3** approaches the male **2**.

(94) In other words, the derivative of a working curve, defined by the extension in the profile plane of the guide cam **124** (having the circumferential coordinate, or angle of rotation of the carousel **10**, in the x-axis and the moulding orientation S in the y-axis) has a first value, corresponding to an initial step of moulding, and a second value, smaller than the first value, corresponding to a final step of moulding.

(95) In an embodiment, the machine **1** comprises an auxiliary unit **14**. The auxiliary unit **14** is configured to allow feeding a charge of plastic material to the female **3** of each mould **11**. The auxiliary unit **14** is configured to allow the closure **100** to be withdrawn from the male **2** of each mould **11**.

(96) In an embodiment, the auxiliary unit **14** comprises a first rotary device **14A**.

(97) In an embodiment, the machine **1** comprises an extruder **15**, configured to extrude a predetermined charge of material to be conveyed into a respective mould.

(98) The first rotary device **14A** comprises a rotary disc **141A** which rotates about a second axis of rotation R2, parallel to the first axis of rotation R1. The rotary disc **141A** is disposed between the upper platen **10A** and the lower platen **10B**. In other words, the rotary disc **141A** is aligned with the upper platen **10A** and the lower platen **10B** along the moulding orientation S. The thickness of the rotary disc **141A** is therefore smaller than the gape SB.

(99) The first rotary device **14A** comprises a plurality of conveyors **142A**. The plurality of conveyors **142A** are connected to the rotary disc **141A** on the underside surface **141A'** thereof, preferably at the edge of the underside surface **141A'**, to rotate as one with the rotary disc **141A**. The conveyors **142A** of the plurality are angularly spaced in such a way that as the first rotary device **14A** rotates, each conveyor is aligned with a respective mould **11** of the machine **1** along the moulding orientation S. The conveyors **142A** each comprise a respective conveying seat **142A'**, configured to receive and hold the charge of material during the rotation of the rotary device **14A**. Each conveyor **142A** is movable between a withdrawing position, where it is out of alignment with the respective mould **11** and aligned with the extruder **15**, along the moulding orientation S, to receive the charge, and a releasing position, where it is aligned with the respective mould **11** (preferably aligned with the female **3** of the respective mould **11**) to release the charge of material.

(100) In an embodiment, the first rotary device **14A** comprises a withdrawal crown **143A**. The withdrawal crown **143A** is configured to withdraw the closures **100** from the male **2** of each mould.

(101) The withdrawal crown **143A** comprises a profiled outer circumference including a plurality of recesses **143A'**, preferably having a semi-circular profile.

(102) The withdrawal crown **143A** is connected to a top surface **141A''** of the rotary disc **141A** and rotates as one therewith. As the withdrawal crown **143A** rotates, each recess **143A'** moves between a withdrawing position, in which it is aligned with a respective mould **11** (preferably with the male **2** of a respective mould **11**) along the moulding orientation **S** in order to withdraw the moulded closure **100**, and a releasing position where it is out of alignment with the respective mould **11** in order to release the moulded closure **100**.

(103) In an embodiment, each recess **143A'** is aligned with a respective conveyor **142A** along the moulding orientation **S**. That way, the conveyor **142A** is aligned with a female **3** of a mould **11** along the moulding orientation **S** at the same time as the recess **143A'** is aligned with the male **2** of the same mould **11**. The machine **1** can therefore withdraw the moulded closure **100** and simultaneously feed the charge of material for the next production cycle.

(104) In an embodiment, the auxiliary unit **14** comprises a second rotary device **14B**. In an embodiment, the rotary device **14B** comprises a conveying crown **141B**, having a respective profiled outer circumference including a respective plurality of recesses **1416'**. When the withdrawal crown **143A** is at its releasing position, the rotary device **14B** is configured to receive from it a moulded closure **100** withdrawn from a respective mould **11**. More in detail, the withdrawal crown **143A** is configured to release the closures **100** withdrawn by it to the conveying crown **141B**. In still more detail, each recess **143A'** of the withdrawal crown **143A** is configured to be aligned radially with a respective recess **141B** of the withdrawal crown **141B** for the latter to convey the corresponding closure **100**.

(105) In an embodiment, the machine **1** comprises a conveyor belt **16**. The second rotary device **14B** is configured to withdraw the closures from the first rotary device **14A** and to release them onto the conveyor belt **16**. More specifically, the conveying crown **141B** is aligned with the conveyor belt along the moulding orientation **S** in order to release the closures **100**.

(106) It should be noted that the mould **11** described in this document has an extension along the moulding orientation **S** that is greater than the average of the moulds that can be used in the machine **1**. For this reason, the gape **SB** of the machine **1** may be too large for conventional moulds (this would also require adapting the stroke of the female **3** relative to the male **2**).

(107) Moreover, with a view to replacing the mould **11** on existing machines, the gape may not be large enough. It is therefore of fundamental importance to provide a solution that makes the machine flexible and capable of working with both kinds of moulds.

(108) This disclosure provides an adapting system configured to vary a minimum distance between the male **2** and the female **3** of each mould **11** along the moulding orientation **S** to allow the auxiliary unit to perform the operations needed to feed the charge of material and withdraw the moulded closure.

(109) In an embodiment, the adapting system is a compression system configured to compress the female **3** of each mould **11**, at the first working station **SL**, that is to say, when the female **3** receives the charge of material and/or when the male **2** releases the moulded closure **100**.

(110) The compression system is configured to compact the first block **31**, the second block **32** and the third block **33** in such a way as to limit the extension of the female **3** of the mould **11**.

(111) In an embodiment, the adapting system is defined by an adaptation performed by the guide cam **124**. In other words, the working curve has a minimum absolute value that takes into account the need for the female **3** to be further down when it is at the first working station **SL1**.

(112) In an embodiment, the male **2** of each mould is made movable relative to the upper platen **10A**. In such an embodiment, therefore, the male **2** of each mould **11** is movable between a withdrawing position, where it is disposed at a first level along the moulding orientation **S** to allow

the auxiliary unit **14** to withdraw the closure **100**, and a moulding position, where it is disposed at a second level along the moulding orientation S, lower than the first level. More specifically, the male **2** is at the withdrawing position when it is at the first working station SL1.

(113) According to an aspect of it, this disclosure also provides a method for making closures, preferably closures for oil containers.

(114) The method comprises a step of preparing a machine **1** for moulding closures from a charge of plastic material. The machine **1** comprises a rotary carousel **10** which rotates about a first axis of rotation R1 and comprising an upper platen **10A** and a lower platen (disc, cylinder) **10B**.

(115) The method comprises a step of preparing a plurality of moulds **11**, each including a male die element **2**, hereinafter denoted simply by the term “male” **2**, and female die element **3**, hereinafter denoted simply by the term “female” **3**, which are aligned along a moulding orientation S, parallel to the first axis of rotation R1. The males **2** of the plurality of moulds are disposed on an edge of the upper platen, angularly spaced from each other. The females **3** of the plurality of moulds are disposed on an edge of the lower platen, angularly spaced from each other.

(116) The method comprises a step of rotating the rotary carousel. During the step of rotating, the rotary carousel **10** transports the plurality of moulds connected to it to the following working stations: a first working station SL1, in which each female **3** receives a charge of material and/or in which each male **2** releases a closure made in a preceding cycle (one complete turn of the rotary carousel **10**); a second working station SL2, in which the female **3** is moved towards and into contact with the male **2** to define the expansion chamber CE and to form the closure; the second working station SL2 being angularly spaced from the first working station SL1 by a closing angle AC; a third working station SL3, in which the female **3** moves away from the male **2** to allow the moulded closure to be released (or withdrawn); the third working station SL3 being angularly spaced from the first working station SL1 by a reopening angle AA.

(117) In an embodiment, the method comprises a step of feeding. The step of feeding is performed preferably when the mould is at the first working station SL1. In the step of feeding, an auxiliary unit **14** feeds a charge of material to the mould **11** which is located at the first working station SL1.

(118) In the step of feeding, a first rotary device **14A** of the auxiliary unit receives a charge of material from an extruder and feeds it to the female **3** of the mould **11** which is located at the first working station SL1. In the step of feeding, a plurality of conveyors **142A** receive respective pluralities of charges of material which are then conveyed to respective females **3** of the moulds **11**.

(119) In an embodiment, the method comprises a step of closing.

(120) The step of closing is performed preferably when the mould is at the second working station SL2.

(121) In the step of closing, the mould moves from an open configuration CA, where the respective male **2** and the respective female **3** are spaced apart, to a closed configuration CC, where the respective male **2** and the respective female **3** are in contact to define the expansion chamber CE.

(122) In an embodiment, in the step of closing, each female **3** moves along the moulding orientation S relative to the male **2** until coming into contact therewith.

(123) In an embodiment, the step of closing comprises a step of closing at the top, in which the male **2** moves relative to the female **3**. More specifically, in the step of closing, the toothed bush **22** and/or the central block **24** move relative to the outer bush **21** from a release position, in which the receiving bush is equidistant or further from the female **3** than the outer bush **21**, to allow the moulded closure **100** to be released, to a moulding position, in which the receiving bush is closer to the female **3** than the outer bush **21**.

(124) The step of closing comprises a step of primary closing. The step of closing comprises a step of secondary closing.

(125) In an embodiment, the step of closing comprises a step of tertiary closing.

(126) In the step of primary closing, a first block **31** of the female **3** moves until it comes into contact with the male **2**. In an embodiment, in the step of primary closing, a second block **32** moves

together with the first block **31** along the moulding orientation S in a closing direction VC (directed from the female **3** to the male **2**).

(127) In this embodiment, therefore, the first block **31** and the second block **32** move until they come into contact with the male **2**.

(128) In the step of secondary closing, a third block **33** moves relative to the first block **31** and/or relative to the second block **32** along the moulding orientation S in the closing direction until coming into contact with the first block **31** and/or the second block **32**. By the step of secondary closing, the expansion chamber CE is completely delimited in order to define the final shape of the closure **100**.

(129) In some advantageous embodiments, the step of secondary closing is a movement of the second block **32** along the moulding orientation S relative to the first block **31** after the first block **31** has already come into contact with the male **2** in the step of primary closing. In these embodiments, the method therefore comprises the step of tertiary closing, in which the third block **33** moves relative to the first block **31** and/or relative to the second block **32** along the moulding orientation S in the closing direction until coming into contact with the first block **31** and/or the second block **32** to define the expansion chamber CE.

(130) For clarity, it is specified that the step of tertiary closing corresponds to the step of secondary closing of the embodiment in which the first block **31** and the second block **32** are constrained to move together in the closing direction VC. The step of tertiary closing is, however, defined only in the case where the first block **31** and the second block **32** are not constrained to move together in the closing direction VC.

(131) The step of closing thus causes the charge of material to migrate into the spaces of the expansion chamber CE defined by the contact between the male **2** and the female **3**.

(132) The fact that the step of closing is modular facilitates migration of the charge of material and improves the quality of spreading the plastic material.

(133) The method comprises a step of holding to allow the plastic material inside the mould to set and harden. In the step of holding, the mould **11** is held in the closed configuration CC along the stretch between the second working station SL2 and the third working station SL3 (that is to say, for a length of time equal to the difference between the opening angle AA and the closing angle AC divided by an angular speed of rotation of the carousel **10**).

(134) The method comprises a step of opening the mould **11**. The step of opening the mould **11** comprises a step of primary opening. The step of opening comprises a step of secondary opening. The step of opening comprises a step of tertiary opening.

(135) In the step of primary opening, the third block **33** of the female **3** is detached from the second block **32** along the moulding orientation S in the opening direction VA. In the step of primary opening, the plunger **331** of the female **3** (of the third block **33**) is detached from the central block **24** of the male **2**.

(136) In the step of secondary opening, the second block **32** moves relative to the first block **31** along the moulding orientation S in the opening direction VA, preferably keeping a constant distance from the third block **33** (in other words, it moves relative to the first block **31** as one with the third block **33**).

(137) In the step of secondary opening, the first collar **101A'**, which is interposed between the first block **31** and the second block **32** with the outer thread FS facing the first block **31**, allows the second block **32** to slide relative to the first block **31** because the thread keeps the first block **31** stationary. Then, when the second block **32** has moved far enough along the moulding orientation S for it to be clear of the first collar **101A'**, the latter can bend in such a way that the first block **31**, too, can move back under the action of the first spring **313** (thus performing the step of tertiary opening).

(138) In the step of tertiary opening, the first block **31** is detached (moves away) from the male **2** along the moulding orientation S in the opening direction VA. In a step of actuating, the first block

31, as it moves, reduces its distance from the second block **32**, until coming into contact therewith. In other embodiments, the first block **31**, as it moves, remains at a constant distance from the second block **32** (in other words, the whole of the female **3** moves as one along the moulding orientation S in the opening direction VA). In an embodiment, the first block **31** moves under the action of the elastic force applied by the first spring **313**, compacting it on the second block **32**.

(139) In an embodiment, the step of opening comprises a step of releasing (detaching). In the step of releasing, the closure **100** just moulded is released from the respective male **2**.

(140) In an embodiment, the step of releasing comprises a step of releasing the fins. In the step of releasing the fins, the central block **24** moves relative to the receiving bush **23** along the moulding orientation S between a working position, in which the central block **24** is proximal to the receiving bush **23** to define the gap **234**, and a detached position, in which the central block **24** is distal from the receiving bush **23** to release the plurality of fins **104**.

(141) More specifically, a piston of the central block **24** (that is, a piston attached to the central block) slides inside a slide cavity **244** of the receiving bush **23**.

(142) In the step of releasing the fins, a pressurizing duct **243** (which, in one embodiment, is also the injection duct **251** of the first cooling circuit **25** of the male **2**) pressurizes an actuating chamber of the slide cavity **244** (the actuating chamber being preferably located upstream of the piston of the central block **24** along the sliding orientation S in the closing direction VC).

(143) The pressure in the actuating chamber acts on the piston of the central block **24** and displaces it from the working position to the detached position.

(144) It should be noted that the pressurizing action of the actuating chamber is only partly responsible for displacing the central block **24**. In effect, the displacement of the central block from the working position to the detached position is due to the movement of the extractor **21** (outer bush **21**). This movement causes the displacement of the closure **100** which, however, on account of the plurality of fins **104**, entrains the central block **24** towards the detached position, overcoming the force of a return spring **242**. Pressurizing the actuating chamber, however, facilitates detachment of the closure by contributing to displacing the central block **24** towards the detached position.

(145) In an embodiment, the step of releasing the fins comprises a step of elastically returning. In the step of elastically returning, the return spring **242** applies an elastic force on the central block **24** (that is, on the piston of the central block **24**) along the moulding orientation S in the opening direction VA. The return spring **242**, being disposed inside the slide cavity **244** at a position downstream of the piston of the central block **24** along the sliding orientation S in the closing direction VC, pushes the central block towards the working position. Thus, when the actuating chamber ceases to be pressurized, the return spring **242** keeps the central block **24** at the working position.

(146) In an embodiment, the step of releasing comprises a step of mutual movement between a moulding unit, which includes the receiving bush **23** and/or the toothed bush **22** and/or the central block **24**, and the outer bush **21**. More specifically, in the step of releasing, the outer bush **21** moves relative to the moulding unit between a moulding position, in which the moulding unit is closer to the female **3** than the outer bush **21**, and a releasing position, in which the moulding unit is equidistant or further from the female **3** than the outer bush **21**, in order to allow the moulded closure **100** to be released. As it moves relative to the moulding unit, the outer bush **21** pushes the connecting collar **101B'** in the closing direction VC, thereby detaching the closure **100** from the moulding unit.

(147) In other embodiments, in the step of moving, the moulding unit moves relative to the outer bush **21**.

(148) The method comprises a step of withdrawing, in which the auxiliary unit **14** picks up (withdraws) the moulded closures **100** from the plurality of moulds **11**.

(149) In the step of withdrawing, a withdrawal crown **143A** of the first rotary device **14A**

withdraws the closures from the male **2** of each mould. The withdrawal crown **143A** is connected to a top surface **141A''** of the rotary disc **141A** and rotates as one therewith.

(150) On a profiled outer circumference of it, the withdrawal crown **143A** comprises a plurality of recesses **143A'**, preferably having a semi-circular profile. The step of withdrawing comprises a step of rotating the withdrawal crown **143A** in which each recess **143A'** moves between a withdrawing position, in which it is aligned with a respective mould **11** (preferably with the male **2** of a respective mould **11**) along the moulding orientation **S** in order to withdraw the moulded closure **100**, and a releasing position where it is out of alignment with the respective mould **11** in order to release the moulded closure **100**.

(151) In an embodiment of the method, each recess **143A'** is aligned with a respective conveyor **142A** along the moulding orientation **S** to perform the step of withdrawing simultaneously with the step of feeding of the next production cycle.

(152) In an embodiment, the method comprises a step of conveying. In the step of conveying, a second rotary device **14B** of the auxiliary unit **14** is configured to convey the closures **100** from the first rotary device **14A** to a further conveyor or to a container. When the withdrawal crown **143A** is at its releasing position, the rotary device **14B** receives from it a moulded closure **100** withdrawn from a respective mould **11**. More in detail, a withdrawal crown **143A** of the first rotary device **14A** releases the closures **100** withdrawn by it to a conveying crown **141B** of the second rotary device **14B**.

(153) In an embodiment, in the step of conveying, the second rotary device **14B** withdraws the closures from the first rotary device **14A** and releases them onto the conveyor belt **16**. More specifically, the conveying crown **141B** is aligned along the moulding orientation **S** and releases the closures **100** onto the conveyor belt **16** by gravity.

(154) It should be noted that the step of conveying by the second rotary device **14B** is optional and might not form part of the method that this disclosure intends to protect.

(155) The method comprises a step of actuating by means of an actuating unit **12**.

(156) In the step of actuating, the rotary carousel **10** is set in rotation about the first axis of rotation **R1**. In the step of actuating, each female **3** (or each male **2**) is actuated so as to move along the moulding orientation **S**, in a closing direction, to close the corresponding mould **11**, or in an opening direction to open the corresponding mould **11**.

(157) The step of actuating comprises a step of setting the rotary carousel in rotation about the first axis of rotation **R1** by means of a rotary actuator **121**.

(158) The step of actuating comprises a step of moving each female **3** (and/or each male **2**) along a moulding orientation **S** by means of a plurality of moulding actuators **122**.

(159) In one of these embodiments, the step of moving each female **3** relative to the male **2** (relative to the frame **1'** of the machine) is performed by a transmission unit which transmits and/or converts the rotary motion of the rotary carousel **10** into a translational motion of a part of the mould **11**, preferably the female **3**.

(160) In this embodiment of the method, the machine **1** comprises, for each female **11**, a respective rod **123A** and a respective slide element **1238**. The rod **123A** is connected to the corresponding female at one end of it and to the slide element **1238** at the opposite end of it.

(161) In this embodiment of the method, a guide cam **124** of the machine **1** guides the slide element **1238** in its movement along the moulding orientation **S**, thus moving each female **3** towards or away from the corresponding male **2**. The step of moving follows a working curve, defined by the extension in the profile plane of the guide cam **124** and having the circumferential coordinate, or angle of rotation of the carousel **10**, in the x-axis and the moulding orientation **S** in the y-axis. That way, the profile of the guide cam **124** defines, for each angular position of the rotary carousel **10**, a corresponding position of the female **3** along the moulding orientation **S**.

(162) The method comprises a step of controlling, in which a control unit **13**, sends drive signals **131'** to the actuating unit **12** (or to the plurality of moulding actuators **122**, or to the rotary actuator

121) in order to drive it. The method comprises a step of setting working parameters, in which a user, through a user interface **131** connected to the control unit **13**, sets the values of the working parameters as a function of which the control unit **13** generates the drive signals **131'** for the actuating unit **12**.

(163) In an embodiment, the method comprises a step of cooling. In the step of cooling, the plurality of moulds **11** are cooled. In the step of cooling, each male **2** and each female **3** are cooled.

(164) The step of cooling comprises a step of cooling the female **3**. In the step of cooling the female **3**, a second cooling circuit **34** preferably cools the first block **31** and/or the second block **32** and/or the third block **33** of the female **3**. In the step of cooling the female **3**, a recirculation duct **34'** of the second cooling circuit **34** circulates a cooling fluid in a cooling direction through the first block **31** and/or the second block **32** and/or the third block **33**. More specifically, in the step of cooling the female **3**, the recirculation duct **34** first cools the third block **33**, then the second block **32** and, lastly, the first block **31**.

(165) Purely by way of example, in the step of cooling the female **3**, the recirculation duct follows the following paths, in chronological order: a first stretch, passing through the plunger **331** along the moulding orientation S in the closing direction VC; a second stretch, passing through the plunger **331** along the moulding orientation S in the opening direction VA; a third stretch which connects one end of the second stretch to the cooling outlet **343B** of the third block **33**; a cooling ring AR, formed on the outside of the first bush **321** of the second block **32** and connected to the cooling inlet **342A** and to the cooling outlet **342B** of the second block **32**; a cooling chamber CR, formed on the outside of the inner bush **314** of the first block **31** and connected to the cooling inlet **341A** and to the cooling outlet **341B** of the first block **31**.

(166) In the step of cooling, the male **2** is cooled by a first cooling circuit **25**. More specifically, an injection duct **251** conveys a cooling fluid to a zone proximate to an annular duct defined between the central block **24** and the receiving bush **23**. In addition, a recycling duct **252** recycles the cooling fluid from the annular duct.

(167) In an embodiment, the method comprises a step of changeover, in which the plurality of moulds **11** are replaced with moulds of different size and features to make closures of a different kind. In the step of changeover, with an equal value of gape SB, the different size of the moulds **11** along the moulding orientation when mounted on the machine **1** varies the distance between the male **2** and the female **3** in the open configuration CA of the mould **11**. Ease of access for the auxiliary unit **14** to feed the charge of material and withdraw the closure **100** is therefore varied.

(168) To overcome these problems, the method comprises a step of adapting.

(169) According to one aspect of the step of adapting, a compression system compresses the female **3** of each mould **11**, at the first working station SL1.

(170) In other words, the compression system compresses the female in order to reduce its dimensions when the female **3** is receiving the charge of material and/or when the male **2** is releasing the moulded closure **100**.

(171) The compression system compacts the first block **31**, the second block **32** and the third block **33** in such a way as to limit the extension of the female **3** of the mould **11** along the moulding orientation S.

(172) According to a further aspect, the step of adapting comprises a step of adapting the guide cam **124** or of replacing the guide cam **124**. Whichever the case, be it adapting or replacing, the working curve has a minimum absolute value that takes into account the need for the female **3** to be further down when it is at the first working station SL1. The minimum absolute value of the working curve will be proportional to the extension of the mould **11** along the moulding orientation S.

(173) The step of adapting might also include a step of moving the male **2** relative to upper platen **10A**. In such an embodiment, therefore, the male **2** of each mould **11** is moved between a withdrawing position, where it is disposed at a first level along the moulding orientation S to allow the auxiliary unit **14** to withdraw the closure **100**, and a moulding position, where it is disposed at a

second level along the moulding orientation S, lower than the first level.

(174) In an embodiment, the step of moving is performed when the male **2** is at the first working station **SL1**.

(175) The following paragraphs, listed in alphanumeric order for reference, are non-limiting example modes of describing this invention.

(176) A. A mould **11** for the production of closures (preferably of plastic material) with a tear-off membrane, in a compression moulding machine **1**, the mould comprising a male die element **2**, including a first abutment surface **21A**, and a female die element **3**, movable relative to each other along a moulding orientation S, such that the mould **11** is positionable between a closed configuration, in which the male die element **2** and the female die element **3** are in contact with each other, and an open configuration, in which the male die element **2** and the female die element **3** are spaced apart, wherein the female die element **3** includes: a first block **31**, a second block **32** and a third block **33**, aligned with each other and movable relative to each other along the moulding orientation S between a position where the first block **31** and the second block **32** are in contact and a position where they are spaced apart and where the second block **32** and the third block **33** are movable relative to each other along the moulding orientation S between a position where they are in contact and a position where they are spaced apart.

(177) A1. The mould according to paragraph A, comprising a first spring **313**, interposed between the first block **31** and the second block **32**, and a second spring **323**, interposed between the second block **32** and the third block **33**.

(178) A1.1. The mould according to paragraph A1, wherein the second spring is configured to apply a closing force **FAL** along the moulding orientation S to keep the second block **32** and the third block **33** spaced apart when no other forces are applied.

(179) A1.2. The mould according to paragraph A1 or A1.1, wherein the first spring **313** is configured to apply a closing force **FAV** along the moulding orientation S to keep the second block **32** and the first block **31** in contact with each other when no other forces are applied.

(180) A2. The mould according to paragraph A1, A1.1 or A1.2, wherein, relative to the moulding orientation S, the first block **31** is interposed between the male die element **2** and the second block **32**, and the second block **32** is interposed between the first block **31** and the third block **33**.

(181) A3. The mould according to any one of paragraphs A to A2. comprising a first cooling circuit **25**, configured to cool the male die element **2**.

(182) A4. The mould according to any one of paragraphs A to A3. comprising a second cooling circuit **34**, configured to cool the female die element **3**.

(183) A4.1. The mould according to paragraph A4, wherein the second cooling circuit **34** comprises a recirculation duct **34'** configured to traverse the first block **31**, the second block **32** and the third block **33**.

(184) A4.1.1. The mould according to paragraph A4.1, wherein the second cooling circuit **34** comprises: a first internal duct **341**, defining a respective cooling path inside the first block **31**; a second internal duct **342**, defining a respective cooling path inside the second block **32**; a third internal duct **343**, defining a respective cooling path inside the third block **33**.

(185) A4.1.1.1. The mould according to paragraph A4.1.1, wherein the first internal duct **341**, the second internal duct **342** and the third internal duct **343** are connected to each other in series.

(186) A4.1.1.2. The mould according to paragraph A4.1.1, wherein the first internal duct **341**, the second internal duct **342** and the third internal duct **343** are connected to each other in parallel.

(187) A5. The mould according to any one of paragraphs A to A4.1.1.2, wherein the male die element **2** includes an annular body at one end of the male die element **2**, facing the female die element **3** and centrally defining a slide cavity **244**.

(188) A5.1. The mould according to paragraph A5, wherein the male die element **2** includes a central block **24**, movable in the slide cavity **244** along the moulding orientation S, between a moulding position, where it is retracted relative to the female die element **3**, and a releasing

position, where it is extended towards the female die element **3**.

(189) A5.1.1. The mould according to paragraph A5.1, wherein the male die element **2** includes a pressurizing duct **243**, connected to the slide cavity **244** and configured to inject air into the slide cavity **244**.

(190) A5.1.1.1. The mould according to paragraph A5.1.1, wherein the annular body comprises a toothed bush **22** and a receiving bush **23**, spaced apart by a flow gap.

(191) A5.1.1.1.1. The mould according to paragraph A5.1.1.1, wherein the annular body comprises at least one communication passage which is open onto the flow gap and onto the pressurizing duct **243** to place them in fluid communication.

(192) A6. The mould according to any one of paragraphs from A to A5.1.1.1.1, wherein the male die element **2** comprises an outer bush **21** that is movable relative to the annular body along the moulding orientation S.

(193) B. A compression moulding machine **1** for making a closure **100** (preferably of plastic material) with a tear-off membrane, comprising: a frame **1'**; a rotary carousel **10** configured to rotate about a first axis of rotation R**1** and including a first platen **10A** and a second platen **10B**; a plurality of moulds **11**, each including a male die element **2**, associated with the first platen **10A**, and a female die element **3**, associated with the second platen **10B**; each mould **11** being positionable between a closed configuration, in which the male die element **2** and the female die element **3** are in contact with each other, and an open configuration, in which the male die element **2** and the female die element **3** are spaced apart; an actuating unit **12**, configured to rotate the rotary carousel and to move each female die element **3** relative to the corresponding first portion **2** along a moulding orientation S; a feeder, configured to provide a charge of material to be moulded; an auxiliary unit **14**, configured to feed the charge to each mould **11** of the plurality of moulds, configured to withdraw the closure from each mould **11** of the plurality of moulds **11** and configured to convey the closure **100** to a storage zone, wherein each mould **11** of the plurality of moulds is a mould **11** according to any one of paragraphs from A to A6.

(194) B1. The machine according to paragraph B, comprising: a first working station SL**1**, in which the auxiliary unit is configured to feed the charge of material to each mould **11**; a second working station SL**2**, in which the actuating unit **12** is configured to move each mould **11** from the open configuration to the closed configuration, the second working station SL**2** being angularly spaced from the first working station SL**1** by a closing angle AC; a third working station SL**3**, in which the actuating unit **12** is configured to move each mould **11** from the closed configuration to the open configuration, the third working station SL**3** being angularly spaced from the first working station SL**1** by a reopening angle AA.

(195) B1.1. The machine according to paragraph B1, wherein, in the first working station SL**1**, the auxiliary unit **14** is configured to withdraw the closure from each mould **11**.

(196) B1.2. The machine according to paragraph B1 or B1.1, comprising a compactor, configured to keep the first block **31**, the second block **32** and the third block **33** of each mould **11** in contact with each other along the moulding orientation S.

(197) B1.2.1. The machine according to paragraph B1.2, wherein the compactor is configured to keep the first block **31**, the second block **32** and the third block **33** of each mould **11** in contact with each other along the moulding orientation S at positions of the mould **11** between the third working station SL**3** and the second working station SL**2**.

(198) B1.2.1.1. The machine according to paragraph B1.2.1 wherein the compactor is configured to keep the first block **31**, the second block **32** and the third block **33** of each mould **11** in contact with each other in the first working station SL**1**.

(199) B2. The machine according to any one of paragraphs from B to B1.2.1.1, wherein the third block **33** is connected to the actuating unit to move the entire female die element **3**.

(200) C. A method for making a closure **100** (preferably of plastic material) with a tear-off membrane, by compression moulding, comprising the following steps: preparing a mould **11**,

having a male die element **2** and a female die element **3**, the female die element **3** including: a first block **31**, a second block **32** and a third block **33**, aligned with each other and movable along a moulding orientation S; feeding a charge of material to the mould **11**; closing the mould **11** by moving the male die element **2** and the female die element **3** of the mould **11** closer together along the moulding orientation S; extruding the charge of material into an expansion chamber CE delimited by the walls of the male die element **2** and of the female die element **3**; opening the mould **11** by moving the male die element **2** and the female die element **3** of the mould **11** away from each other along the moulding orientation S; withdrawing the closure **100** from the mould **11**.

(201) C1. The method according to paragraph C, wherein a first spring **313**, interposed between the first block **31** and the second block **32**, applies a first force along the moulding orientation S, between the first block **31** and the second block **32**, and wherein a second spring **323**, interposed between the second block **32** and the third block **33**, applies a second force along the moulding orientation S, between the second block **32** and the third block **33**.

(202) C1.1. The method according to paragraph C1, wherein the first force is a closing force FAV along the moulding orientation S to keep the first block **31** and the second block **32** in contact with each other when no other forces are applied.

(203) C1.2. The method according to paragraph C1 or C1.1, wherein the first force is an opening force FAL along the moulding orientation S to keep the second block **32** and the third block **33** spaced apart when no other forces are applied.

(204) C1.3. The method according to any one of paragraphs C1 to C1.2, wherein the step of closing comprises a first step of closing in which, as the male die element **2** and the female die element **3** move closer together, the first block **31** and the second block **32** are in contact with each other under the effect of the first spring and the third block **33** is spaced from the second block **32** under the effect of the second spring **323**.

(205) C1.3.1. The method according to paragraph C1.3, wherein the step of closing comprises a second step of closing, in which, when the second block **32** comes into abutment against the male die element **2** along the moulding orientation S, the third block **33** continues moving until coming into contact with the second block **32**.

(206) C2. The method according to any one of paragraphs C to C1.3.1, wherein the step of opening comprises a first step of opening in which the third block **33** moves away from the second block **32** along the moulding orientation S.

(207) C2.1. The method according to paragraph C2, wherein the step of opening comprises a second step of opening in which the second block **32** moves away from the first block **31** along the moulding orientation S while keeping constant its distance from the third block **33** along the moulding orientation S.

(208) C2.1.1. The method according to paragraph C2.1, wherein the step of opening comprises a third step of opening in which the first block **31** moves away from the male die element **2** as one with the second block **32** and the third block **33**.

(209) C2.1.1.1 The method according to paragraph C2.1.1, wherein the first step of opening is performed before the second step of opening and the second step of opening is performed before the third step of opening.

(210) C3. The method according to any one of paragraphs C to C2.1.1, comprising a first step of cooling, in which a first cooling circuit **25** of the mould **11** cools the male die element **2**.

(211) C4. The method according to any one of paragraphs C to C3, comprising a second step of cooling, in which a second cooling circuit **34** of the mould **11** cools the female die element **3**.

(212) C5. The method according to any one of paragraphs C to C4, wherein the step of releasing comprises a step of sliding a central block **24** of the male die element **2** along the moulding orientation S relative to a slide cavity **244** formed in an annular body of the male die element **2**.

(213) C4.1. The method according to paragraph C4, wherein the step of releasing comprises a step of sliding an outer bush **21** of the male die element **2** along the moulding orientation S relative to

the annular body in order to release the closure **100**.

(214) C5. The method according to any one of paragraphs C to C4.1, comprising the following steps: preparing a rotary carousel **10** on which the mould **11** is mounted; performing the step of feeding the charge of material in a first working station SL1 by means of an auxiliary unit **14**; rotating the rotary carousel **10**; performing the step of closing the mould in a second working station SL2 of the rotary carousel **10** by means of an actuating unit **12**; performing the step of opening the mould in a third working station SL3 of the rotary carousel **10** by means of the actuating unit **12**; performing the step of withdrawing the closure **100** from the mould **11** in the first working station SL1 by means of the auxiliary unit **14**.

(215) C5.1. The method according to paragraph C5, comprising a step of compacting, in which a compactor reduces the height of the female die element **3** along the moulding orientation S.

(216) C5.1.1. The method according to paragraph C5.1, wherein, in the step of compacting, the compactor keeps the second block **32** in contact with the third block **33** during the steps of feeding and withdrawing.

(217) C6. The method according to any one of paragraphs C to C5.1.1, wherein, when the mould is closed, the second block and the first block are kept in contact with each other, when no other forces are applied.

(218) C6.1. The method according to paragraph C6, wherein, when the mould is closed, the second block and the first block are kept in contact with each other by applying a closing force.

(219) C6.1.1. The method according to paragraph C6.1, wherein closing force is provided by a first spring.

(220) C6.1.1. The method according to paragraph C6.1.1, wherein the first spring applies the closing force along the moulding orientation.

Claims

1. A method for making a closure of plastic material with a tear-off membrane, by compression moulding, comprising the following steps: preparing a mould, having a male die element and a female die element, the female die element including: a first block, a second block and a third block, aligned with each other and movable relative to each other along a moulding orientation; closing the mould by moving the male die element and the female die element closer together along the moulding orientation to compress a charge of material so as to form the closure; opening the mould by moving the male die element and the female die element away from each other along the moulding orientation to allow withdrawing the closure from the mould, and wherein, while the mould is being opened, the first block moves away from the second block and the second block moves away from the third block, wherein, when the mould is closed, the first block and the second block are in contact with each other, wherein, when the mould is closed, the second block and the first block are kept in contact with each other by applying a closing force, when no other forces are applied.

2. The method according to claim 1, wherein, when the mould is closed, a first spring applies a closing force along the moulding orientation to keep the second block and the first block in contact with each other when no other forces are applied.

3. The method according to claim 2, wherein the step of closing comprises a first step of closing, in which, as the male die element and the female die element move closer together, the first block and the second block are in contact with each other under the effect of the first spring and the third block is spaced from the second block under the effect of a second spring, which applies an opening force along the moulding orientation between the second block and the third block to keep them spaced apart when no other forces are applied.

4. The method according to claim 3, wherein the step of closing further comprises a second step of closing, in which, when the second block comes into abutment against the male die element along

the moulding orientation, the third block continues moving until coming into contact with the second block.

5. The method according to claim 1 comprising a step of cooling, in which a first cooling circuit of the mould cools the male die element and a second cooling circuit of the mould cools the female die element.

6. The method according to claim 1, comprising a step of releasing the closure from the mould, the step of releasing including the following sub-steps: moving a central block of the male die element along the moulding orientation towards the female die element in a slide cavity formed in an annular body of the male die element; moving an outer bush of the male die element along the moulding orientation towards the female die element relative to the annular body.

7. The method according to claim 6, wherein the step of releasing comprises a step of sliding an outer bush of the male die element along the moulding orientation relative to the annular body in order to release the closure.

8. The method according to claim 1, comprising the following steps: preparing a rotary carousel on which the mould is mounted; feeding the charge of material to the mould in a first working station; performing the step of closing the mould in a second working station of the rotary carousel; performing the step of opening the mould in a third working station of the rotary carousel; withdrawing the closure from the mould in the first working station.

9. The method according to claim 8, comprising a step of compacting in which a compactor keeps the first block in contact with the second block and the second block in contact with the third block during the steps of feeding and withdrawing.

10. The method according to claim 1, wherein closing force is provided by a first spring.

11. The method according to claim 10, wherein the first spring applies the closing force along the moulding orientation.

12. The method according to claim 1, wherein the step of opening comprises a first step of opening in which the third block moves away from the second block along the moulding orientation.

13. The method according to claim 12, wherein the step of opening comprises a second step of opening in which the second block moves away from the first block along the moulding orientation while keeping constant its distance from the third block along the moulding orientation.

14. A method for making a closure of plastic material with a tear-off membrane, by compression moulding, comprising the following steps: preparing a mould, having a male die element and a female die element, the female die element including: a first block, a second block and a third block, aligned with each other and movable along a moulding orientation; feeding a charge of material to the mould; closing the mould by moving the male die element and the female die element of the mould closer together along the moulding orientation; extruding the charge of material into an expansion chamber delimited by the walls of the male die element and of the female die element; opening the mould by moving the male die element and the female die element of the mould away from each other along the moulding orientation; withdrawing the closure from the mould, wherein a first spring, interposed between the first block and the second block, applies a first force along the moulding orientation, between the first block and the second block, and wherein a second spring, interposed between the second block and the third block, applies a second force along the moulding orientation, between the second block and the third block.

15. The method according to claim 14, wherein the first force is a closing force along the moulding orientation to keep the first block and the second block in contact with each other when no other forces are applied.

16. The method according to claim 14, wherein the first force is an opening force along the moulding orientation to keep the second block and the third block spaced apart when no other forces are applied.

17. The method according to claim 14, wherein the step of closing comprises a first step of closing in which, as the male die element and the female die element move closer together, the first block

and the second block are in contact with each other under the effect of the first spring and the third block is spaced from the second block under the effect of the second spring.

18. The method according to claim 17, wherein the step of closing comprises a second step of closing, in which, when the second block comes into abutment against the male die element along the moulding orientation, the third block continues moving until coming into contact with the second block.

19. A method for making a closure of plastic material with a tear-off membrane, by compression moulding, comprising the following steps: preparing a mould, having a male die element and a female die element, the female die element including: a first block, a second block and a third block, aligned with each other and movable relative to each other along a moulding orientation; closing the mould by moving the male die element and the female die element closer together along the moulding orientation to compress a charge of material so as to form the closure; opening the mould by moving the male die element and the female die element away from each other along the moulding orientation to allow withdrawing the closure from the mould, and wherein, while the mould is being opened, the first block moves away from the second block and the second block moves away from the third block, wherein, when the mould is closed, the first block and the second block are in contact with each other, wherein, when the mould is closed, a first spring applies a closing force along the moulding orientation to keep the second block and the first block in contact with each other when no other forces are applied.

20. A method for making a closure of plastic material with a tear-off membrane, by compression moulding, comprising the following steps: preparing a mould, having a male die element and a female die element, the female die element including: a first block, a second block and a third block, aligned with each other and movable relative to each other along a moulding orientation; closing the mould by moving the male die element and the female die element closer together along the moulding orientation to compress a charge of material so as to form the closure; opening the mould by moving the male die element and the female die element away from each other along the moulding orientation to allow withdrawing the closure from the mould, and wherein, while the mould is being opened, the first block moves away from the second block and the second block moves away from the third block, wherein, when the mould is closed, the first block and the second block are in contact with each other, comprising a step of releasing the closure from the mould, the step of releasing including the following sub-steps: moving a central block of the male die element along the moulding orientation towards the female die element in a slide cavity formed in an annular body of the male die element; moving an outer bush of the male die element along the moulding orientation towards the female die element relative to the annular body.
